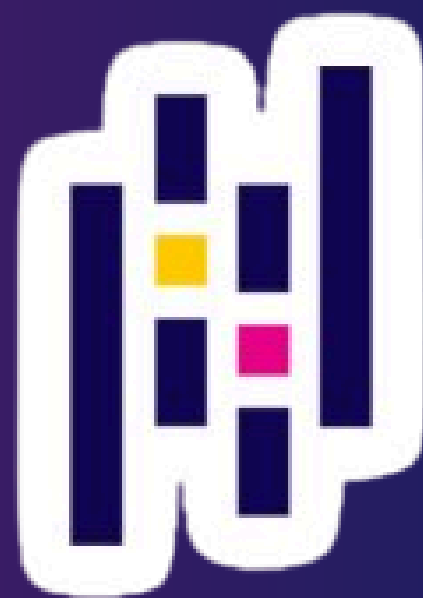
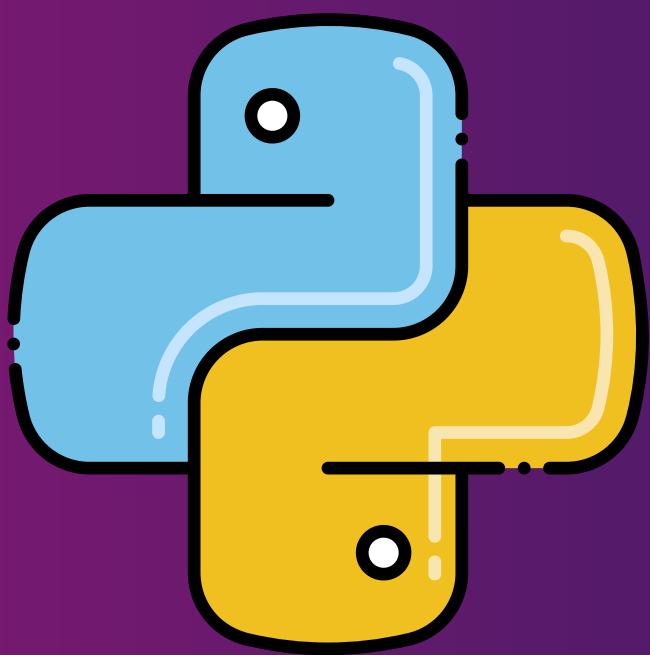


Pandas for AI

Data Manipulation Guide

Turn messy data into machine learning ready datasets.



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Why Pandas for AI?



Pandas makes it clean

- Load any dataset
- Clean efficiently
- Transform easily



Raw data is messy

- Missing values
- Wrong formats
- Inconsistent structure

80% of ML work is data prep. Pandas handles it all. Built on NumPy. Made for data.



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Loading Datasets



```
import pandas as pd

# Read CSV file
df = pd.read_csv('data.csv')

# Read Excel
df = pd.read_excel('data.xlsx')

# Read JSON
df = pd.read_json('data.json')

# View first 5 rows
df.head()

# View last 5 rows
df.tail()

# Dataset shape
df.shape # (rows, columns)
```



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Quick Data Exploration



```
# Basic info
```

```
df.info() # Data types, non-null counts
```

```
# Statistical summary
```

```
df.describe() # Mean, std, min, max
```

```
# Column names
```

```
df.columns
```

```
# Data types
```

```
df.dtypes
```

```
# Check for missing values
```

```
df.isnull().sum()
```

```
# Unique values in column
```

```
df['column_name'].unique()
```



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Selecting Data



```
# Select single column
```

```
df['age']
```

```
# Select multiple columns
```

```
df[['name', 'age', 'city']]
```

```
# Select rows by index
```

```
df.iloc[0]          # First row
```

```
df.iloc[0:5]        # First 5 rows
```

```
# Select by condition
```

```
df[df['age'] > 25]
```

```
# Multiple conditions
```

```
df[(df['age'] > 25) & (df['city'] ==  
'NYC')]
```



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Data Cleaning

Remove missing values

```
df.dropna() # Drop rows with any NaN
```



Fill missing values

```
df.fillna(0) # Replace NaN with 0
```

```
df.fillna(df.mean()) # Fill with mean
```

Remove duplicates

```
df.drop_duplicates()
```

Rename columns

```
df.rename(columns={'old_name':  
'new_name'})
```

Change data type

```
df['age'] = df['age'].astype(int)
```



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Data Transformation



```
# Create new column
```

```
df['total'] = df['price'] *  
df['quantity']
```

```
# Apply function to column
```

```
df['age_squared'] =  
df['age'].apply(lambda x: x**2)
```

```
# Sort data
```

```
df.sort_values('age', ascending=False)
```

```
# Group and aggregate
```

```
df.groupby('city')['sales'].mean()
```

```
# Reset index
```

```
df.reset_index(drop=True)
```



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Filtering & Statistics



```
# Filter rows
```

```
adults = df[df['age'] >= 18]
```

```
# Count values
```

```
df['city'].value_counts()
```

```
# Aggregate functions
```

```
df['sales'].sum()
```

```
df['sales'].mean()
```

```
df['sales'].max()
```

```
df['sales'].min()
```

```
# Group by and aggregate
```

```
df.groupby('category').agg({  
    'sales': 'sum',  
    'quantity': 'mean'  
})
```



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Combining DataFrames

Concatenate (stack)

```
pd.concat([df1, df2])
```



Merge (like **SQL join**)

```
pd.merge(df1, df2, on='id')
```

Left join

```
pd.merge(df1, df2, on='id', how='left')
```

Inner join (default)

```
pd.merge(df1, df2, on='id', how='inner')
```

Append rows

```
df1.append(df2)
```

Export cleaned data

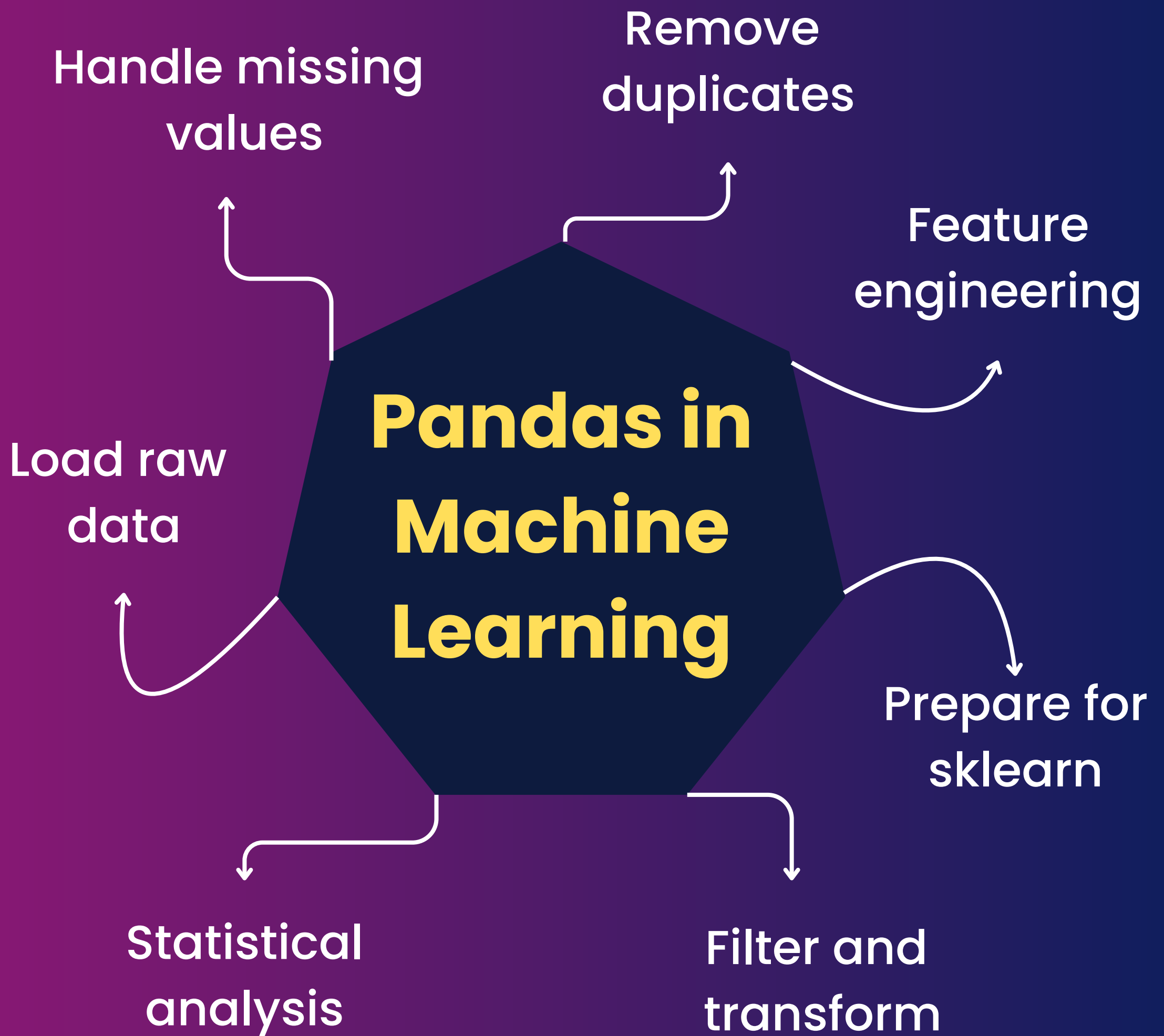
```
df.to_csv('clean_data.csv', index=False)
```



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Pandas in Machine Learning



Every ML Project Starts with Pandas



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